

RENT CAR PROCESSING SYSTEM

LAB GROUP P3

1 INTRODUCTION

In this project, the lab group will design and implement a complex Web application for desktop and mobile devices. The project is divided into three modules. Each module addresses a different project segment and will be worked on by 1 to 3 teams (the number of teams required for each module is specified below). The teams working in each module will focus on a different portion. When integrating, the different teams' work will form a working module, and the three modules should come together to make the entire application work.

The Rent Car Ltd is a car rental company located at the airport. Rent car has a fleet of cars ranging from small compact cars to luxurious cars. The company has the capability to wash, repair and servicing their cars. From time to time, the company will also sell their retired cars.

Rent Car Ltd provides car rentals to locals and international visitors with a valid driving licence for more than two years and driver age 22 or older. An additional deposit of S\$200 is required for rentals with driving licence less than two years or driver is under 22 years old. The driving licence validation is usually done as an in-process when the customer comes to pick up the vehicle.

Rent Car Ltd staff complained to the CEO about the inaccuracy of the cars available for rental. Moreover, staff are not happy with the current manual system as it is time consuming in term of placing and processing rental orders. However, finance office staff see the current system is acceptable and meets their needs. Customers also have their own concerns related to booking their rental cars and instructions to pick up and return the cars.

The process to pick up the car is, a staff will look up for the booking details and confirm with the customer if the details are correct. Then, the staff will validate all named drivers in the booking licence(s). Finally, the final rental cost and deposit (starting from S\$300) are collected from the customer before the vehicle is issued. After issuing the car the customer will check the car with a staff and sign off as the handover process.

Whenever, a car returns from rental, it will undergo inspection for damages, damages recorded, repairs (if any), servicing and cleaning. The returning process should about three hours to complete before the car is placed back for rentals again. Parts of this process is not tracked at the moment.

Rent Car Ltd is not the only company in the market and its competitor "Yellow Rentals" company is growing dramatically in the last three years, therefore Rent Car's CEO requested to implement a new information management system that provide a quick and efficient service. The new system

should be able to process a customer's rental faster, more efficient, and track the process more accurately.

Section 2 below describes the Web application, known as the "Rent Car Processing System", which allows Rent Car's customers, staff, and executive team to obtain accurate information about the rental, services, and market trends. The Web application is divided into THREE (3) modules: (1) Rental Processing System, (2) Inventory Management System and (3) In & Out Processes. Each team should be able to complete component(s) of their assigned module over the course of the trimester.

Modules will be assigned to teams by the first lab. You are expected to discuss with other teams in your lab, and especially team(s) assigned to the same module to agree on the boundary and component(s) that each team will work on. We recommend that you do this as early as you can to avoid delays to your work. You will find the module and the component(s) you are working on and their boundaries gets clearer as you progress through your project and with further analysis of the requirements.

Each team member should identify at least one user feature to design and develop within their module. All team members must own their feature(s), and no two features should overlap more than 40%. Every team within the module should work on separate parts of the module. This means no overlap in functionality within the module. We have listed several requirements to get you started, however you are free (and should) add on to ensure your work scope meets the project requirements.

2 RENT CAR PROCESSING SYSTEM (RCPS)

The Rent Car Processing System (RCPS) is a Web application, which manages the inventory and rental requests. The Web application is divided into THREE (3) modules for engineering purposes.

Module 1: Rental Processing System (up to three teams)

The new rental processing system must be able to track the booking from the time it is received until the car is issued and returned. Customers must be able to place a booking and view the status of their bookings. Additionally, customers is able to cancel a booking up to two hours before the car is pick up.

All rental bookings must be done through the Web RCPS, even for walk-in customers. Staff of Rent Car Ltd will be in-store to assist customers queries. Hence, the customer must be able to browse through the available cars get a quote for the rental online. The details of each car, including the car colour, the interior seating details (with pictures) and boot capacity (with pictures) must be shown to the customer.

To begin the searching for car rental process, the customer must first enter the rental dates and pick up and return time, and the system will automatically identify the available cars. You will need to work with Module 2: Inventory Management System, to retrieve these details.

The system main functions are place order, query and order, order status notification, orders dispatched, order returns, returning process status, and deposit refunds. To place an order online, a valid customer id will be required. Walk-in customers details will be filled in on behalf by a Pro Rental's staff.

After selecting the car to rent, customers will be able to specific the number of drivers. An additional charge of S\$20 will be include for every additional driver named. All rentals must purchase insurance. Customers can choose the type of insurance coverage:

- Basic covers up to 100km per rental is S\$10 per 24 hours
- Basic Plus covers up to 150km per rental is S\$15 per 24 hours
- Essentials covers up to 120km per rental + chips is S\$20 per 24 hours
- Essential Plus covers up to 200km per rental + chips + collision damage waiver (CDW) is S\$32 per 24 hours
- Peace of Mind covers unlimited kilometer per rental and everything covered in Essential Plus is S\$35 per 24 hours

The structure and prices for the insurance can be updated by a staff as require from time to time to align with the market pricing. This feature must be managed by this module.

The availability of cars is managed by Module 2 – Inventory Management System. The base value of the car rental per day is determined by the staff conducting the In-process checks in Module 3 - In & Out Processes.

The system should upsell the customer to take on addons like Sat Nav (S\$15 per rental), child booster seat (S\$35 per rental), fragrance (S\$2 per rental), and umbrellas (S\$3 per rental).

Finally, the system should tally the total rent amount quote. Customers can choose to pay full upfront at a 2% discount (no refund) or a holding fee of S\$50 and the remainder at pick up at full price (cancellation more than 2 hours, the holding fee is returned).

Customers can sign up to notification of their rental so that they would be reminded of their rentals and returning of the vehicle. The system must provide a dashboard to deal with customers rental details and deposit refunds.

All customer's details and login credentials are managed fully by this module too.

Module 2: Inventory Management System (two teams)

The Car Rent Ltd requires a smart management of their inventory that includes a dashboard for the company's staff to manage their fleet, retire cars, purchasing new cars and addon equipment (e.g., Sat Nav, child booster seat etc.). The system holds all the details of their cars, categorises of the cars, and the stock level in a centralised database.

Adding, updating, deleting, searching, returning, displaying and viewing of cars and equipment are the main functions of the system.

Other functions such as keep track of inventory levels, alert when stocks run low for certain category and season are recommended. It is important for the company to identifying cars and equipment that are popular, so provide a prediction for the items two months ahead, especially for holiday and festive seasons. It is important to highlight cars and equipment that are **not moving for more than two years** so that the company can take actions to clear these items. Purchasing of new and high demand items are only done through **trusted business sources**.

The system must **maintain a list of trusted business sources** that they can purchase their cars and equipment with. These trusted sources are categories into three groups: Long credit period; Quick turnaround time; and New untested.

Whenever a car is returned from rental, a return process will be triggered and the status of the stage where the returned car and equipment must be made known to Rent Car's staff and the customer's dashboard. The return process will go through the following stages.

1. Damage Inspection and re-valuation
2. Repairing (optional step – only applicable for damaged items)
3. Servicing
4. Cleaning
5. Return to inventory for rental or for retirement sales.

The clearing of cars and equipment that are not moving for more than two years are done through sales in the physical store. Once identified, these items should be shifted to a different list, done by a staff. This new list should record when the item is sold and the recommended and final price sold.

The system will manage all the daily transactions and generate the required reports easily. Having a centralised database helps data integrity. In addition, the system should have a graphical user interface for easy interaction and to reduce staff training time.

The company is keen to develop a smart system to cope with market changes and competition with other companies such as "Yellow Rentals". Therefore, the management team requested that the system should provide reports showing analysis and visual presentation that give an insight about the data to help managers make decisions in the right time.

Module 3: In & Out Processes (two teams)

This module covers four main stages.

- (1) Check out process: when the customer comes to pick up the vehicle.

There should be a checklist of things to cover for the car and add-on equipment. At the end of this process, the customer must sign off with the date and time to agree with the assessment. This entire process should be digitalised and recorded in the system for module 2 processes.

- (2) Return process: when the customer returns the vehicle.

When the car is returned, the customer will drop off the keys in the Keys Box and notify a staff. The staff will pick up the key to begin the Return process stage.

Similar a checklist of thing to cover should be provided digitally.

- 1) Record return date/time (to calculate late fees).
- 2) Record ending mileage (to calculate overage charges).
- 3) Record ending fuel level (to calculate refuelling charges).
- 4) Log new damages found during inspection.
- 5) Proceed to stage 3 - Repairing, servicing and cleaning process
- 6) Proceed to stage 4 – Conduct valuation assessment
- 7) Assign the car to return to inventory for rental.

- (3) Repairing, servicing and cleaning process

All repairs to the car and equipment must be recorded with the cost. If no repair is required, process to service the car. There should be a digitalised list of things to do. E.g., pump tyres, check engine oil etc. These processes, measurements, and items used (including price) must be recorded.

Finally, the cleaning process will take place. Again, a digital list of things to do should be provided and checked off.

All the checked off items time and date should be recorded to.

Besides repairing, the time taken for the servicing and cleaning process should be recorded and should not exceed 1.5 hours. The system to alert the staff if the time exceeds.

- (4) Valuation assessment

A staff will inspect the equipment and car's condition and past records to determine the current value of the equipment and car. The system should provide the staff with the necessary details to make the decision. The staff will then enter the current valuation of the staff into the system.

It is also at this point that the staff will indicate whether a car or equipment should return into inventory for rental or should retire the car or equipment. There will be cases when it is a total write off. These decision should be recorded in the system with necessary remarks for the decision if it is a retirement or write off.